

-- ===== DIMENSIONS =====

```
CREATE TABLE dw.DimGeography (  
    GeographyKey INT NOT NULL,  
    GeographyType NVARCHAR(50) NOT NULL,  
    ContinentName NVARCHAR(50) NOT NULL,  
    CityName NVARCHAR(100) NULL,  
    StateProvinceName NVARCHAR(100) NULL,  
    RegionCountryName NVARCHAR(100) NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_DimGeography PRIMARY KEY (GeographyKey)  
);
```

```
CREATE TABLE dw.DimCurrency (  
    CurrencyKey INT NOT NULL,  
    CurrencyLabel NVARCHAR(10) NOT NULL,  
    CurrencyName NVARCHAR(20) NULL,  
    CurrencyDescription NVARCHAR(50) NOT NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_DimCurrency PRIMARY KEY (CurrencyKey)  
);
```

```
CREATE TABLE dw.DimDate (  
    DateKey INT NOT NULL,  
    DateLabel NVARCHAR(10) NOT NULL,  
    DateName NVARCHAR(20) NULL,  
    DateDescription NVARCHAR(50) NOT NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_DimDate PRIMARY KEY (DateKey)  
);
```

Datekey DATETIME NOT NULL,
FullDateLabel NVARCHAR(20) NOT NULL,
DateDescription NVARCHAR(20) NOT NULL,
CalendarYear INT NOT NULL,
CalendarYearLabel NVARCHAR(20) NOT NULL,
CalendarHalfYear INT NOT NULL,
CalendarHalfYearLabel NVARCHAR(20) NOT NULL,
CalendarQuarter INT NOT NULL,
CalendarQuarterLabel NVARCHAR(20) NULL,
CalendarMonth INT NOT NULL,
CalendarMonthLabel NVARCHAR(20) NOT NULL,
CalendarWeek INT NOT NULL,
CalendarWeekLabel NVARCHAR(20) NOT NULL,
CalendarDayOfWeek INT NOT NULL,
CalendarDayOfWeekLabel NVARCHAR(10) NOT NULL,
FiscalYear INT NOT NULL,
FiscalYearLabel NVARCHAR(20) NOT NULL,
FiscalQuarter INT NOT NULL,
FiscalQuarterLabel NVARCHAR(20) NOT NULL,
IsWorkDay NVARCHAR(20) NOT NULL,
IsHoliday INT NOT NULL,
HolidayName NVARCHAR(20) NOT NULL,
LoadDate DATETIME NULL,
PackageName NVARCHAR(128) NULL,
ETLUser NVARCHAR(100) NULL,
BatchID NVARCHAR(50) NULL,
CONSTRAINT PK_DimDate PRIMARY KEY (Datekey)
);

```
CREATE TABLE dw.DimProductCategory (  
    ProductCategoryKey INT NOT NULL,  
    ProductCategoryLabel NVARCHAR(100) NULL,  
    ProductCategoryName NVARCHAR(30) NOT NULL,  
    ProductCategoryDescription NVARCHAR(50) NOT NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_DimProductCategory PRIMARY KEY (ProductCategoryKey)  
);
```

```
CREATE TABLE dw.DimProductSubcategory (  
    ProductSubcategoryKey INT NOT NULL,  
    ProductSubcategoryLabel NVARCHAR(100) NULL,  
    ProductSubcategoryName NVARCHAR(50) NOT NULL,  
    ProductSubcategoryDescription NVARCHAR(100) NULL,  
    ProductCategoryKey INT NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_DimProductSubcategory PRIMARY KEY (ProductSubcategoryKey),  
    CONSTRAINT FK_DimProductSubcategory_DimProductCategory FOREIGN KEY (ProductCategoryKey)  
        REFERENCES dw.DimProductCategory (ProductCategoryKey)  
);
```

```
CREATE TABLE dw.DimProduct (  
    ProductKey INT NOT NULL,
```

```
ProductLabel NVARCHAR(255) NULL,  
ProductName NVARCHAR(500) NULL,  
ProductDescription NVARCHAR(400) NULL,  
ProductSubcategoryKey INT NULL,  
Manufacturer NVARCHAR(50) NULL,  
BrandName NVARCHAR(50) NULL,  
ColorName NVARCHAR(20) NOT NULL,  
Size NVARCHAR(50) NULL,  
Weight FLOAT NULL,  
UnitOfMeasureName NVARCHAR(40) NULL,  
StockTypeName NVARCHAR(40) NULL,  
UnitCost MONEY NULL,  
UnitPrice MONEY NULL,  
AvailableForSaleDate DATETIME NULL,  
PriceTier NVARCHAR(20) NULL,  
ProductAvailability NVARCHAR(20) NULL,  
LoadDate DATETIME NULL,  
PackageName NVARCHAR(128) NULL,  
ETLUser NVARCHAR(100) NULL,  
BatchID NVARCHAR(50) NULL,  
CONSTRAINT PK_DimProduct PRIMARY KEY (ProductKey),  
CONSTRAINT FK_DimProduct_DimProductSubcategory FOREIGN KEY (ProductSubcategoryKey)  
    REFERENCES dw.DimProductSubcategory (ProductSubcategoryKey)  
);
```

```
CREATE TABLE dw.DimStore (  
    StoreKey INT NOT NULL,  
    GeographyKey INT NULL,  
    StoreLabel NVARCHAR(200) NULL,
```

```
StoreName NVARCHAR(100) NOT NULL,
StoreDescription NVARCHAR(200) NULL,
StoreManager NVARCHAR(200) NULL,
StoreType NVARCHAR(50) NULL,
StoreManagementType NVARCHAR(50) NULL,
AddressLine1 NVARCHAR(200) NULL,
ZipCode NVARCHAR(20) NULL,
StorePhone NVARCHAR(20) NULL,
StoreFax NVARCHAR(50) NULL,
EmployeeCount INT NULL,
SellingAreaSize INT NULL,
WeeklyHours FLOAT NULL,
OpenDate DATETIME NULL,
CloseDate DATETIME NULL,
CloseReason NVARCHAR(200) NULL,
StoreOperationalStatus NVARCHAR(20) NULL,
StoreSizeCategory NVARCHAR(20) NULL,
LoadDate DATETIME NULL,
PackageName NVARCHAR(128) NULL,
ETLUser NVARCHAR(100) NULL,
BatchID NVARCHAR(50) NULL,
CONSTRAINT PK_DimStore PRIMARY KEY (StoreKey),
CONSTRAINT FK_DimStore_DimGeography FOREIGN KEY (GeographyKey)
    REFERENCES dw.DimGeography (GeographyKey)
);
```

```
CREATE TABLE dw.DimCustomer (
    CustomerKey INT NOT NULL,
    GeographyKey INT NULL,
```

```
CustomerLabel NVARCHAR(200) NULL,  
FirstName NVARCHAR(50) NULL,  
LastName NVARCHAR(50) NULL,  
BirthDate DATETIME NULL,  
MaritalStatus NVARCHAR(1) NULL,  
Gender NVARCHAR(1) NULL,  
EmailAddress NVARCHAR(50) NULL,  
AddressLine1 NVARCHAR(200) NULL,  
Phone NVARCHAR(20) NULL,  
Occupation NVARCHAR(100) NULL,  
HouseOwnerFlag NVARCHAR(1) NULL,  
NumberCarsOwned TINYINT NULL,  
NumberChildrenAtHome TINYINT NULL,  
TotalChildren TINYINT NULL,  
Education NVARCHAR(40) NULL,  
YearlyIncome MONEY NULL,  
IncomeSegment NVARCHAR(20) NULL,  
CustomerLifeStage NVARCHAR(20) NULL,  
LoadDate DATETIME NULL,  
PackageName NVARCHAR(128) NULL,  
ETLUser NVARCHAR(100) NULL,  
BatchID NVARCHAR(50) NULL,  
CONSTRAINT PK_DimCustomer PRIMARY KEY (CustomerKey),  
CONSTRAINT FK_DimCustomer_DimGeography FOREIGN KEY (GeographyKey)  
    REFERENCES dw.DimGeography (GeographyKey)  
);
```

```
CREATE TABLE dw.DimChannel (  
    ChannelKey INT NOT NULL,
```

```

ChannelLabel NVARCHAR(100) NOT NULL,
ChannelName NVARCHAR(20) NOT NULL,
LoadDate DATETIME NULL,
PackageName NVARCHAR(128) NULL,
ETLUser NVARCHAR(100) NULL,
BatchID NVARCHAR(50) NULL,
CONSTRAINT PK_DimChannel PRIMARY KEY (ChannelKey)
);

```

```

CREATE TABLE dw.DimPromotion (
    PromotionKey INT NOT NULL,
    PromotionLabel NVARCHAR(200) NULL,
    PromotionName NVARCHAR(100) NOT NULL,
    PromotionDescription NVARCHAR(255) NULL,
    DiscountPercent FLOAT NULL,
    PromotionType NVARCHAR(50) NULL,
    PromotionCategory NVARCHAR(50) NULL,
    StartDate DATETIME NULL,
    EndDate DATETIME NULL,
    PromotionStatus NVARCHAR(20) NULL,
    DiscountTier NVARCHAR(20) NULL,
    LoadDate DATETIME NULL,
    PackageName NVARCHAR(128) NULL,
    ETLUser NVARCHAR(100) NULL,
    BatchID NVARCHAR(50) NULL,
    CONSTRAINT PK_DimPromotion PRIMARY KEY (PromotionKey)
);

```

```

-- ===== FACTS =====

```

```
CREATE TABLE dw.FactOnlineSales (  
    OnlineSalesKey INT NOT NULL,  
    DateKey DATETIME NOT NULL,  
    StoreKey INT NOT NULL,  
    ProductKey INT NOT NULL,  
    PromotionKey INT NOT NULL,  
    CurrencyKey INT NOT NULL,  
    CustomerKey INT NOT NULL,  
    SalesOrderNumber NVARCHAR(50) NOT NULL,  
    SalesOrderLineNumber INT NULL,  
    SalesQuantity INT NOT NULL,  
    SalesAmount MONEY NOT NULL,  
    ReturnQuantity INT NULL,  
    ReturnAmount MONEY NULL,  
    DiscountQuantity INT NULL,  
    DiscountAmount MONEY NULL,  
    TotalCost MONEY NULL,  
    UnitCost MONEY NULL,  
    UnitPrice MONEY NULL,  
    GrossProfit MONEY NULL,  
    NetSalesAmount MONEY NULL,  
    ProfitMarginPercent NUMERIC(30,4) NULL,  
    AverageUnitProfit NUMERIC(30,15) NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_FactOnlineSales PRIMARY KEY (OnlineSalesKey),
```



```
CONSTRAINT FK_FactOnlineSales_DimDate FOREIGN KEY (DateKey)
    REFERENCES dw.DimDate (Datekey),
CONSTRAINT FK_FactOnlineSales_DimStore FOREIGN KEY (StoreKey)
    REFERENCES dw.DimStore (StoreKey),
CONSTRAINT FK_FactOnlineSales_DimProduct FOREIGN KEY (ProductKey)
    REFERENCES dw.DimProduct (ProductKey),
CONSTRAINT FK_FactOnlineSales_DimPromotion FOREIGN KEY (PromotionKey)
    REFERENCES dw.DimPromotion (PromotionKey),
CONSTRAINT FK_FactOnlineSales_DimCurrency FOREIGN KEY (CurrencyKey)
    REFERENCES dw.DimCurrency (CurrencyKey),
CONSTRAINT FK_FactOnlineSales_DimCustomer FOREIGN KEY (CustomerKey)
    REFERENCES dw.DimCustomer (CustomerKey)
);
```

```
CREATE TABLE dw.FactSales (
    SalesKey INT NOT NULL,
    DateKey DATETIME NOT NULL,
    ChannelKey INT NULL,
    StoreKey INT NOT NULL,
    ProductKey INT NOT NULL,
    PromotionKey INT NULL,
    CurrencyKey INT NOT NULL,
    UnitCost MONEY NULL,
    UnitPrice MONEY NULL,
    SalesQuantity INT NOT NULL,
    ReturnQuantity INT NULL,
    ReturnAmount MONEY NULL,
    DiscountQuantity INT NULL,
    DiscountAmount MONEY NULL,
```

```
TotalCost MONEY NULL,
SalesAmount MONEY NOT NULL,
GrossProfit MONEY NULL,
NetSalesAmount MONEY NULL,
ProfitMarginPercent NUMERIC(30,4) NULL,
AverageUnitProfit NUMERIC(30,15) NULL,
LoadDate DATETIME NULL,
PackageName NVARCHAR(128) NULL,
ETLUser NVARCHAR(100) NULL,
BatchID NVARCHAR(50) NULL,
CONSTRAINT PK_FactSales PRIMARY KEY (SalesKey),
CONSTRAINT FK_FactSales_DimDate FOREIGN KEY (DateKey)
    REFERENCES dw.DimDate (Datekey),
CONSTRAINT FK_FactSales_DimChannel FOREIGN KEY (ChannelKey)
    REFERENCES dw.DimChannel (ChannelKey),
CONSTRAINT FK_FactSales_DimStore FOREIGN KEY (StoreKey)
    REFERENCES dw.DimStore (StoreKey),
CONSTRAINT FK_FactSales_DimProduct FOREIGN KEY (ProductKey)
    REFERENCES dw.DimProduct (ProductKey),
CONSTRAINT FK_FactSales_DimPromotion FOREIGN KEY (PromotionKey)
    REFERENCES dw.DimPromotion (PromotionKey),
CONSTRAINT FK_FactSales_DimCurrency FOREIGN KEY (CurrencyKey)
    REFERENCES dw.DimCurrency (CurrencyKey)
);
```

```
CREATE TABLE dw.FactInventory (
    InventoryKey INT NOT NULL,
    DateKey DATETIME NOT NULL,
    StoreKey INT NOT NULL,
```

```
ProductKey INT NOT NULL,
CurrencyKey INT NOT NULL,
OnHandQuantity INT NULL,
OnOrderQuantity INT NULL,
SafetyStockQuantity INT NULL,
UnitCost MONEY NULL,
DaysInStock INT NULL,
MinDayInStock INT NULL,
MaxDayInStock INT NULL,
Aging INT NULL,
InventoryValue NUMERIC(30,4) NULL,
OnOrderValue NUMERIC(30,4) NULL,
StockStatus NVARCHAR(20) NULL,
StockCoverageDays INT NULL,
LoadDate DATETIME NULL,
PackageName NVARCHAR(128) NULL,
ETLUser NVARCHAR(100) NULL,
BatchID NVARCHAR(50) NULL,
CONSTRAINT PK_FactInventory PRIMARY KEY (InventoryKey),
CONSTRAINT FK_FactInventory_DimDate FOREIGN KEY (DateKey)
    REFERENCES dw.DimDate (Datekey),
CONSTRAINT FK_FactInventory_DimStore FOREIGN KEY (StoreKey)
    REFERENCES dw.DimStore (StoreKey),
CONSTRAINT FK_FactInventory_DimProduct FOREIGN KEY (ProductKey)
    REFERENCES dw.DimProduct (ProductKey),
CONSTRAINT FK_FactInventory_DimCurrency FOREIGN KEY (CurrencyKey)
    REFERENCES dw.DimCurrency (CurrencyKey)
);
```

```
CREATE TABLE dw.FactExchangeRate (  
    ExchangeRateKey INT NOT NULL,  
    CurrencyKey INT NOT NULL,  
    DateKey DATETIME NOT NULL,  
    AverageRate FLOAT NULL,  
    EndOfDayRate FLOAT NULL,  
    RateVariance FLOAT NULL,  
    RateVariancePercent FLOAT NULL,  
    RateTrend NVARCHAR(12) NULL,  
    LoadDate DATETIME NULL,  
    PackageName NVARCHAR(128) NULL,  
    ETLUser NVARCHAR(100) NULL,  
    BatchID NVARCHAR(50) NULL,  
    CONSTRAINT PK_FactExchangeRate PRIMARY KEY (ExchangeRateKey),  
    CONSTRAINT FK_FactExchangeRate_DimCurrency FOREIGN KEY (CurrencyKey)  
        REFERENCES dw.DimCurrency (CurrencyKey),  
    CONSTRAINT FK_FactExchangeRate_DimDate FOREIGN KEY (DateKey)  
        REFERENCES dw.DimDate (Datekey)  
);
```